Probability Homework #8 (Coverage: 7.1, 7.2, 7.3)　　資工二 113590021 邱冠勛

1. Suppose that 15 points are selected at random and independently from the interval (0, 1). In average, how many of them will be greater than 3/4?

$P(X) = \frac{1 - 3/4}{1 - 0} = \frac{1}{4}$

$X \sim B(15, \frac{1}{4})$

$E(X) = np = 15 \times \frac{1}{4} = \frac{15}{4}$

ANS: $\frac{15}{4}$

2. A line segment of length k is cut randomly into two segments. What is the probability that none of the two segments is smaller than $k/3$?

$P(X) = \frac{\frac{1}{3}k}{k} = \frac{1}{3}$

ANS: $\frac{1}{3}$

3. Let X be a uniform random number from (0, 1). Find the density functions of $Y = -\ln(1 - X)$.

$$F(X) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } 0 \le x < 1 \\ 1 & \text{if } 1 \le x \end{cases}$$

$G(t) = P(Y < t) = P(-\ln(1-X) < t)$　　$t \in (0, \infty)$

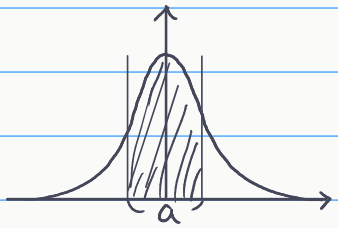
$= P(\ln(1-X) > -t) = P(1 - X > e^{-t})$

$= P(X < 1 - e^{-t})$

$$= \begin{cases} 0 & t \le 0 \\ 1 - e^{-t} & t > 0 \end{cases}$$

ANS: $g(t) = \begin{cases} 0 & t \le 0 \\ e^{-t} & t > 0 \end{cases}$ #

4. Let Z be a standard normal random variable and α be a given positive constant. Find the real number x that maximizes $P(x < Z < x + \alpha)$.



ANS: $-\frac{a}{2}$

5. The grades for a certain exam are normally distributed with mean 67 and variance 64. What percent of students get A($\geq$ 90), B(80 – 90), C(70 – 80), D(60 – 70), and F($<$ 60)?

$\mu = 67 \quad \sigma^2 = 64 \quad \sigma = 8$

A: $1 - P(x \leq 90)$

$= 1 - P\left(\frac{x-\mu}{\sigma} \leq \frac{90-67}{8}\right)$

$= 1 - \Phi\left(\frac{23}{8}\right) \approx 0.00212$ #

B: $P\left(\frac{80-67}{8} \leq z \leq \frac{90-67}{8}\right)$

$= \Phi\left(\frac{23}{8}\right) - \Phi\left(\frac{13}{8}\right)$

$= 0.99788 - 0.94738 = 0.0505$#

C: $P\left(\frac{70-67}{8} \leq z \leq \frac{80-67}{8}\right)$

$= \Phi\left(\frac{13}{8}\right) - \Phi\left(\frac{3}{8}\right)$

$= 0.94738 - 0.64431 = 0.30307$ #

D: $P\left(\frac{60-67}{8} \leq z \leq \frac{70-67}{8}\right)$

$= \Phi\left(\frac{3}{8}\right) - \Phi\left(\frac{-7}{8}\right)$

$= 0.64431 - (1 - 0.80785) = 0.45216$ #

E: $P\left(z \leq \frac{60-67}{8}\right)$

$= \Phi\left(-\frac{7}{8}\right)$

$= 1 - 0.80785 = 0.19215$ #

6. Let $X \sim N(\mu, \sigma^2)$. Prove that $P(|X - \mu| > k\sigma)$ does not depend on μ or σ.

Since μ is the average value of X, so substract it would cause the graph to centerized to x=0, then divide X by σ would turn standard deviation into 1, so this two steps would turn all Normal R.V. into a Standard Normal R.V., thus the probability would be the same since it's already standardized

$P(|X-\mu| > k\sigma) = P\left(\frac{|X-\mu|}{\sigma} > k\right) = P(|z| > k)$

$= P(z > k) + P(z < -k)$

$= 1 - \Phi(k) + (1 - \Phi(k))$

$= 2 - 2\Phi(k)$

↑ not related to μ & σ

7. Let $X \sim N(0,1)$. Calculate the probability density function of $Y = \sqrt{|X|}$.

$F(x) = \Phi(x)$

$G(t) = P(Y < t) = P(\sqrt{|X|} < t) = P(-t^2 < X < t^2)$

$= \Phi(t^2) - \Phi(-t^2)$

$= 2\Phi(t^2) - 1$

$g(t) = G'(t) = 2\phi(t^2) \cdot 2t = \begin{cases} 2\phi(t^2) \cdot 2t & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases}$ *

8. Let X be an exponential random variable with parameter λ, mean $E(X)$ and standard deviation σ_X. Find $P(|X - E(X)| > 2\sigma_X)$.

$E(X) = \frac{1}{\lambda}$ $\quad Var(X) = \frac{1}{\lambda^2}$ $\quad \sigma = \frac{1}{\lambda}$

$P(|X - E(X)| > 2\sigma_X) = P(X < -2\sigma_X + E(X)) + P(X > 2\sigma_X + E(X))$

$= F(-\frac{1}{\lambda}) + (1 - F(\frac{3}{\lambda}))$

$= 0 + 1 - (1 - e^{-3}) = e^{-3}$ #

9. Let X, the lifetime (in years) of a radio tube, be exponentially distributed with mean $1/\lambda$. Find the probability mass function (p.m.f.) of $\lfloor X+1 \rfloor$. (It is noted that $\lfloor X+1 \rfloor$ is the integer part of $X+1$, i.e. the greatest integer less than or equal to $X+1$. Actually, $\lfloor X+1 \rfloor$ can be proved to be a Geometric random variable.)

$P(Y=k) = P(\lfloor X+1 \rfloor = k)$

$= P(k \leq X+1 < k+1)$

$= P(k-1 \leq X < k)$

$= F(k) - F(k-1)$

$= 1 - e^{-\lambda k} - (1 - e^{-\lambda(k-1)})$

$= 1 - e^{-\lambda k} - 1 + e^{-\lambda k} \cdot e^{\lambda}$

$= (e^{\lambda} - 1) e^{-\lambda k}$

$= (e^{\lambda} - 1) e^{-\lambda(k-1)} \cdot e^{-\lambda}$

$= (1 - e^{-\lambda}) \cdot e^{-\lambda(k-1)}$

$= (1-p) \cdot p^{k-1} \quad \cdots \quad p = e^{-\lambda}$ #